

Day 2 of java Class Notes

Data Types in Java

1. **Bytes** – 8 bytes
 2. **Short** – 16bytes
 3. **Int** - 32 bytes
 4. **Long** - 64 bytes
 5. **Char**
 6. **Boolean**
 7. **Float**
 8. **Double**
- **String is class in Java also act as an Datatype**
 - `//we can access attributes directly in class methods but in main method we have to make object of the class in order to access the attributes of the class`
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Strings in JAVA

- **String is a class in Java**
- **It exists in lang package.**
- 1. **toString()**
- 2. **toUpperCase()**
- 3. **toLowerCase()**
- 4. **valueOf(int)**
- 5. **toCharArray()**
- 6. **substring(int)**
- 7. **substring(*beingIndex*:int,*endIndex*:int)** // generally used- use case can be found in string.java file
- **by default if we console object then it returns the garbage value (cuz gc method of lang.System class takes care)like `Strdemo@15db9742` but if we introduce toString in our program then it takes care and return what is returned using it .**
- **next()** reads only one word till encounters space

- `nextLine()` reads wholeline until line changes
- we can only write body of `toString` method , but we cant for rest method as they are predefined.
- Whatever java takes or gives, it takes as string on black screen.
- `String> (s1).equals(s2)` method

print vs println vs printf

Constructor in JAVA

- Constructor is a special member method which is invoked automatically at the time of creation of object of class.
- Types: -
 1. Default Constructor
 2. Parameterized Constructor
- Constructor name is same as of the class name.
- Both type of Constructor could remain in same class and could be used each at a time or simultaneously by creating two different objects.
- During the time of Object creation, it depends on how object is created with param or without param, based on that Default or Parameterized is called.
- There is no return type of Constructor.
- Syntax :-

```
Class A{  
    A(){  
        //Default Constructor  
    }  
    A(int a,int b){  
        //Parameterized Constructor  
    }  
    public static void main(String arg[]){  
        A x = new A(); // this will be automatically called as this is  
        Constructor  
    }  
}
```

Check constructor.java for Example

- `this` is a Keyword which refers to current object
- this() should be used in any one Constructor else it will lead to:

```
PS A:\Coday\Java Winter Traning\Day 2> javac .\constructor.java
.\constructor.java:10: error: recursive constructor invocation
    Constructor(int a,int b){
    ^
1 error
```

- Also this() should be used in first line else it will lead to :

```
PS A:\Coday\Java Winter Traning\Day 2> javac .\constructor.java
.\constructor.java:9: error: call to this must be first statement in constructor
    this(15,16);
    ^
1 error
```

Static Keyword in Java(M.imp)

- Static keyword is used for memory management only.
- It is used with Variable, methods and class.
- Static Variable are loaded in memory at the same time of class loading.
- The Values Static variables are shared among the object of class.

static Block

- Static Block are used to initialize static Variables.
- Runs before Main method.

Syntax:>

```
static{
.....
.....
}
```

```
static int c = 0;
    Staticdemo(){
        c=c+1;
        System.out.println(c);
```

```
}
```

```
int c = 0; // for first set of inputs i.e. 1 , 1, 1
```

```
PS A:\Coday\Java Winter Traning\Day 2> java Staticdemo
1
1
1
PS A:\Coday\Java Winter Traning\Day 2> javac .\staticdemo.java
PS A:\Coday\Java Winter Traning\Day 2> java Staticdemo
1
2
3
PS A:\Coday\Java Winter Traning\Day 2> 
```

- Static → Default Cons → Class Methods [Order of their run].
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